A Central Limit Theorem for High-Temperature SYK Model

Abstract

Fix an even interaction order $q \geq 4$, and let $Z_N(\beta)$ denote the partition function of the Gaussian Sachdev–Ye–Kitaev model with N Majorana fermions. We prove that for all sufficiently small inverse temperatures $\beta > 0$, there exists $\rho = \rho(\beta) > 0$ such that

$$N^{q/2-1}(\log Z_N(\beta) - \mathbb{E}\log Z_N(\beta)) \Longrightarrow \mathcal{N}(0, \rho^2).$$

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1 Introduction and main results

1.1 The SYK model

Fix an even integer $q \geq 4$, and let N be even. Write

$$\mathcal{I}_N = \{I \subset [N] : |I| = q\}, \qquad \sigma_N^2 = \frac{(q-1)!}{N^{q-1}},$$

and let $G = (g_I)_{I \in \mathcal{I}_N}$ be independent standard Gaussian random variables. Let $\psi_1, \ldots, \psi_N$ be Majorana matrices satisfying $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$. For $I = \{i_1 < \cdots < i_q\}$ set

$$\psi_I = \mathrm{i}^{q/2}\psi_{i_1} \cdots \psi_{i_q}.$$

Then ψ_I is Hermitian, $\psi_I^2 = \mathbb{I}$, and $\|\psi_I\| = 1$. Define

$$H_N(G) = \sigma_N \sum_{I \in \mathcal{I}_N} g_I \psi_I, \qquad Z_N(z; G) = \operatorname{tr} e^{-zH_N(G)}, \qquad F_N(z) = \mathbb{E}Z_N(z; G).$$

Here $\operatorname{tr} = 2^{-N/2}\operatorname{Tr}$ denotes the normalized trace; in particular, $Z_N(0; G) = F_N(0) = 1$. This normalization does not affect the relative fluctuations studied below. We sometime write $Z_N(\beta) = Z_N(\beta; G)$ for brevity. Throughout, all limits are taken as $N \to \infty$ through even integers.

1.2 Main results

It is studied the limiting behavior of the logarithmic partition function in high-temperature regime:

Theorem 1.1. *There exists $\tilde{\beta}_q > 0$ such that for any $\beta \in [-\tilde{\beta}_q, \tilde{\beta}_q]$, there exists $\ell_q(\beta)$ such that*

$$\lim_{N\to\infty} N^{-1}\log Z_N = \lim_{N\to\infty} N^{-1}\log \mathbb{E}[Z_N] = \ell_p(\beta) \qquad \textit{a.s.}$$

Moreover, $\ell_p(\beta)$ is differentiable over $(-\tilde{\beta}_q, \tilde{\beta}_q)$.

Our main result is the following fluctuation resutls:

Theorem 1.2. *There exists $\beta_q \in (0, \tilde{\beta}_q]$ such that for any $\beta \in (-\beta_q, \beta_q)$,*

$$N^{q/2-1}\left(\frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)} - 1\right) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2}\beta^2(\ell_q'(\beta))^2\right). \tag{1}$$

Moreover,

$$\frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\sqrt{\operatorname{Var} Z_N(\beta)}} \Longrightarrow \mathcal{N}(0,1). \tag{2}$$

Corollary 1.3. *Under the assumptions of Theorem 1.2,*

$$N^{q/2-1}\left(\log Z_N(\beta) - \mathbb{E}\log Z_N(\beta)\right) \Longrightarrow \mathcal{N}\left(0, \frac{q!}{2}\,\beta^2(\ell_q'(\beta))^2\right).$$

1.3 A brief background and relation to earlier work

The Sachdev–Ye–Kitaev (SYK) model is a mean-field model of randomly interacting fermions originating in the work of Sachdev and Ye [1] and its Majorana formulation by Kitaev. Its combination of solvability, strong interactions, and chaotic behavior has made it a central model in quantum statistical mechanics and holography; see, for example, [2]. At positive temperature, the partition function contains both the thermodynamic information and the sample-to-sample fluctuations of the disorder.

Rigorous results have recently identified the high-temperature annealed and quenched free-energy limits [5] and established complex zero-freeness and second-moment estimates [4]. The purpose of this paper is to determine the leading distributional fluctuations in the same high-temperature regime.

Feng, Tian, and Wei [3] proved a central limit theorem for linear eigenvalue statistics of the Gaussian SYK Hamiltonian. Their theorem treats a fixed test function f under regularity assumptions on f and f'. In the normalization used in that paper, however, the partition function corresponds to

$$f_{N,\beta}(x) = e^{-\beta\tau_N x}, \qquad \tau_N \asymp \sqrt{N},$$

so both the test function and its derivative vary with N. The fixed-test-function theorem therefore does not apply directly to $Z_N(\beta)$.

1.4 Outline of the proof

By applying the chaos expansion, since Clifford sign-gauge symmetry eliminates the first Gaussian chaos and identifies the second chaos exactly, we have the following:

$$Z_N(\beta) = \mathbb{E}Z_N(\beta) + A_N(\beta)Q_N + \mathcal{R}_N(\beta),$$

where

$$A_N(\beta) = \frac{\beta}{2}F_N'(\beta), \qquad Q_N = \frac{1}{|\mathcal{I}_N|}\sum_I (g_I^2 - 1). \tag{3}$$

Here $\mathcal{R}_N$ is the sum of the Gaussian chaoses of orders at least three. Thus the leading random term is a normalized sum of $\mathcal{I}_N$ independent centered variables. The main task is to prove

$$\|\mathcal{R}_N(\beta)\|_2^2 = o(N^{-(q-2)}).$$

Indeed, a Duhamel derivative bound and the sign-gauge parity rule give

$$\sum_{r\geq 3}\|P_r Z_N(\beta)\|_2^2 \leq \sum_{r\geq 3}(\beta\sigma_N)^{2r} \sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!}. \tag{4}$$

Using a combinatorial argument, we have that the entire higher-order sum is bounded by $O(N^{-3(q/2-1)}) = o(N^{-(q-2)})$.

Moreover, the annealed free-energy input and convexity imply

$$\frac{A_N(\beta)}{N\mathbb{E}Z_N(\beta)} \longrightarrow \frac{\beta}{2}\ell_q'(\beta).$$

Since $M_N \sim N^q/q!$, the classical central limit theorem for Q_N, the remainder estimate gives the claimed limit for $Z_N/\mathbb{E}Z_N$.

2 Gaussian-chaos decomposition

We use the probabilists' Hermite polynomials

$$H_\alpha(G) = \prod_{I\in\mathcal{I}_N} \mathrm{He}_{\alpha_I}(g_I), \qquad \mathbb{E}[H_\alpha H_\eta] = \alpha!\,\mathbf{1}_{\{\alpha=\eta\}}, \tag{5}$$

where $\alpha! = \prod_I \alpha_I!$. Since H_α consists of complete orthonormal basis in $L^2(\gamma_{|\mathcal{I}_N|})$ where $\gamma_{|\mathcal{I}_N|}$ denotes standard Gaussian measure on $\mathbb{R}^{|\mathcal{I}_N|}$, we have

$$Z_N(\beta : G) = \sum_{k=0}^{\infty} \sum_{\alpha:|\alpha|=k} \frac{\mathbb{E}[Z_N(\beta : G')H_\alpha(G')]}{\alpha!} H_\alpha(G), \tag{6}$$

where the second sum runs over all multi-index α such that $\sum_I \alpha_I = k$, and the expectation is taken over G' independent copy of G. Let P_r denote the orthogonal projection in $L^2(\gamma_{|\mathcal{I}_N|})$ onto the r-th homogeneous Gaussian chaos: Then, we can write

$$P_r Z_N(G) = \sum_{\alpha:\ |\alpha|=k} \frac{\mathbb{E}[Z_N(\beta : G')H_\alpha(G')]}{\alpha!} H_\alpha(G).$$

For every sufficiently regular random variable X,

$$\mathbb{E}[(P_r X_2)^2] = \sum_{|\alpha|=r} \frac{|\mathbb{E}\,\partial_g^\alpha X|^2}{\alpha!}. \tag{7}$$

Recall $A_N(\beta) := \frac{\beta}{2} F_N'(\beta)$ from (3). The next lemma proves the exact radial decomposition

$$Z_N(\beta) = \mathbb{E}Z_N(\beta) + A_N(\beta)Q_N + \mathcal{R}_N(\beta), \qquad \mathcal{R}_N = \sum_{r\ge 3} P_r Z_N. \tag{8}$$

2.1 Exact finite-size radial projection

Call a multi-index α *even* if

$$\sum_{I\ni x} \alpha_I \equiv 0 \pmod 2 \qquad \text{for every } x \in [N]. \tag{9}$$

Otherwise, we call α odd.

Lemma 2.1. *For $s = (s_1, \dots, s_N) \in \{\pm 1\}^N$, define $\chi_s(I) = \prod_{x\in I} s_x$. We have*

$$Z_N(\beta; G) = Z_N\big(\beta; (\chi_s(I) g_I)_I\big). \tag{10}$$

In particular, if α is odd, then

$$\mathbb{E}[Z_N(\beta : G')H_\alpha(G') = 0.$$

Proof. To make the Clifford symmetry explicit, let

$$\Gamma := \psi_1 \cdots \psi_N.$$

Note that $\Gamma^*\Gamma$ is identity map, so Γ is unitary. Since N is even, moving ψ_j through the other $N-1$ Majorana generators produces the sign $(-1)^{N-1} = -1$. Hence

$$\Gamma \psi_j \Gamma^* = -\psi_j \qquad (j \in [N]).$$

For each $x \in [N]$, put $U_x = \Gamma\psi_x$. This is unitary, and conjugation by U_x is the composition of conjugation by ψ_x and conjugation by Γ. The Clifford relations give

$$\psi_x \psi_j \psi_x = \begin{cases} \psi_x, & j = x, \\ -\psi_j, & j \neq x. \end{cases}$$

Combining the last two displays, we obtain

$$U_x \psi_j U_x^* = \begin{cases} -\psi_x, & j = x, \\ \psi_j, & j \neq x. \end{cases}$$

Thus conjugation by U_x changes the sign of precisely the generator ψ_x.

Let $S_s = \{x \in [N] : s_x = -1\}$ and let U_s be the product of the unitaries U_x over $x \in S_s$, in any fixed order. Successively applying the preceding conjugation rule yields

$$U_s \psi_j U_s^* = s_j \psi_j \qquad (j \in [N]).$$

If $I = \{i_1 < \cdots < i_q\}$, then conjugation is multiplicative, so the definition $\psi_I = \mathrm{i}^{q/2}\psi_{i_1} \cdots \psi_{i_q}$ gives

$$\begin{aligned} U_s \psi_I U_s^* &= \mathrm{i}^{q/2} (U_s \psi_{i_1} U_s^*) \cdots (U_s \psi_{i_q} U_s^*) \\ &= \left(\prod_{x \in I} s_x \right) \psi_I = \chi_s(I) \psi_I. \end{aligned}$$

Writing $g^{(s)} = (\chi_s(I) g_I)_I$, we therefore have

$$\begin{aligned} H_N(g^{(s)}) &= \sigma_N \sum_{I \in \mathcal{I}_N} \chi_s(I) g_I \psi_I \\ &= \sigma_N \sum_{I \in \mathcal{I}_N} g_I U_s \psi_I U_s^* = U_s H_N(G) U_s^*. \end{aligned}$$

The power-series definition of the exponential then implies

$$e^{-\beta H_N(g^{(s)})} = U_s e^{-\beta H_N(G)} U_s^*.$$

Finally, the normalized trace is invariant under unitary conjugation. Consequently

$$Z_N(\beta; g^{(s)}) = \operatorname{tr}\left(U_s e^{-\beta H_N(G)} U_s^* \right) = \operatorname{tr} e^{-\beta H_N(G)} = Z_N(\beta; G),$$

which proves (10).

Assume that α is odd, i.e. we can take x such that $\sum_{I : x \in I} \alpha_I$ is odd. We take s such that $s_y = 1$ for $y \neq x$ and $s_x = 1$. Then,

$$H_\alpha((\chi_s(I) g_I)_I) = \left(\prod_{x=1}^{N} s_x^{\sum_{I \ni x} \alpha_I} \right) H_\alpha(G) = -H_\alpha(G).$$

Hence, we have

$$\mathbb{E}[Z_N(G') H_\alpha(G')] = \mathbb{E}[Z_N(\chi_s G') H_\alpha(\chi_s G')] = -\mathbb{E}[Z_N(G') H_\alpha(G')],$$

which gives the second claim. □

Lemma 2.2 (First and second chaoses)**.** *For every real β,*

$$P_1 Z_N(\beta) = 0$$

and

$$P_2 Z_N(\beta) = \frac{\beta\, F_N'(\beta)}{2M_N} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1). \tag{11}$$

Proof. This parity rule above kills the first chaos. In the second chaos it also kills every off-diagonal monomial $g_I g_J$, since the parity condition for such a monomial forces $I = J$. Permutation symmetry shows that the remaining diagonal coefficients agree. Consequently

$$P_2 Z_N(\beta) = b_N(\beta) \sum_I (g_I^2 - 1)$$

for a deterministic scalar b_N.

It remains to compute the coefficient. For each fixed I, the probabilists' second Hermite polynomial is $g_I^2 - 1$, with

$$\mathbb{E}(g_I^2 - 1)^2 = 2.$$

Since P_2 is an orthogonal projection and the variables g_I are independent, taking the inner product of the preceding formula for $P_2 Z_N$ with $g_I^2 - 1$ gives

$$2b_N(\beta) = \mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big].$$

Applying Gaussian integration by parts twice, conditionally on all coordinates other than g_I, yields

$$\mathbb{E}\big[Z_N(\beta)(g_I^2 - 1)\big] = \mathbb{E}\, g_I \partial_{g_I} Z_N(\beta).$$

Summing over the $\mathcal{I}_N$ choices of I therefore gives the first equality below. A further conditional Gaussian integration by parts gives the second:

$$\begin{aligned} 2M_N b_N(\beta) &= \sum_I \mathbb{E}\big[g_I \partial_{g_I} Z_N(\beta)\big] \\ &= \beta\, \partial_\beta \mathbb{E} Z_N(\beta) = \beta F_N'(\beta). \end{aligned}$$

The last equality follows pointwise from the exact scaling identity

$$Z_N(\beta; tG) = \operatorname{tr} e^{-\beta H_N(tG)} = \operatorname{tr} e^{-t\beta H_N(G)} = Z_N(t\beta; G),$$

where we used $H_N(tG) = tH_N(G)$. Differentiating at $t = 1$ gives

$$\sum_I g_I \partial_{g_I} Z_N(\beta; G) = \beta\, \partial_\beta Z_N(\beta; G).$$

This proves (11). □

2.2 Direct Rademacher control of all higher chaoses

Lemma 2.3. *For real β, every multi-index α of total order r satisfies*

$$\big|\partial_g^\alpha Z_N(\beta; G)\big| \le (|\beta|\sigma_N)^r Z_N(\beta; G). \tag{12}$$

Proof. Label the r differentiations by $I_1, \dots, I_r$. The iterated Duhamel formula is a sum over $r!$ permutations of integrals over the simplex $\Delta_r = \{t_j \ge 0 : \sum_{j=0}^r t_j = 1\}$, whose volume is $1/r!$. Every integrand, apart from the factor $(|\beta|\sigma_N)^r$, has the form

$$\operatorname{tr}\left(e^{-t_0\beta H_N} \psi_{I_{\pi(1)}} e^{-t_1\beta H_N} \cdots \psi_{I_{\pi(r)}} e^{-t_r\beta H_N}\right).$$

Here and below the Schatten norms are defined using the normalized trace:

$$\|A\|_p = (\operatorname{tr}|A|^p)^{1/p}, \qquad \|A\|_\infty = \|A\|.$$

For $t_j > 0$,

$$\left\| e^{-t_j \beta H_N} \right\|_{1/t_j} = \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j},$$

with the usual $p = \infty$ interpretation when $t_j = 0$. Noncommutative Hölder, with Schatten exponents $1/t_j$, and $\|\psi_I\|_\infty = 1$, therefore bounds the absolute value of the integrand by

$$\prod_{j=0}^{r} \left(\operatorname{tr} e^{-\beta H_N}\right)^{t_j} = Z_N(\beta; G).$$

The permutation count cancels the simplex volume, proving (12). The same labeled formula covers repeated coordinates. □

Recall that Lemma 2.1 shows that only even multi-indices have nonzero Hermite coefficients.

Let $(\varepsilon_x)_{x=1}^N$ be independent Rademacher variables on an auxiliary probability space, and write $\mathbb{E}_\varepsilon$ for expectation on this space. Set

$$L_N = \sum_{x=1}^{N} \varepsilon_x, \qquad S_{N,q} = \sum_{I \in \mathcal{I}_N} \varepsilon_I, \qquad \varepsilon_I = \prod_{x \in I} \varepsilon_x. \tag{13}$$

Thus $S_{N,q}$ is the degree-q elementary symmetric polynomial in the Rademacher signs.

Lemma 2.4. *For every integer $r \geq 0$,*

$$\sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!} = \frac{1}{r!} \mathbb{E}_\varepsilon\left[S_{N,q}^r\right]. \tag{14}$$

Moreover, there are constants $c_q > 0$ and $C_q < \infty$, depending only on q, such that

$$\mathbb{E}_\varepsilon\left[|S_{N,q}|^3 e^{t|S_{N,q}|}\right] \leq C_q N^{3q/2} \tag{15}$$

for every N and every $0 \leq t \leq c_q N^{-(q-1)}$.

Proof. Expanding $S_{N,q}^r$ by the multinomial formula gives

$$\frac{1}{r!} \mathbb{E}_\varepsilon\left[S_{N,q}^r\right] = \sum_{|\alpha|=r} \frac{1}{\alpha!} \mathbb{E}_\varepsilon\left[\prod_{I \in \mathcal{I}_N} \varepsilon_I^{\alpha_I}\right]. \tag{16}$$

For a fixed α, the product inside the expectation is

$$\prod_{x=1}^{N} \varepsilon_x^{\sum_{I \ni x} \alpha_I}. \tag{17}$$

Its expectation is one if α is even and zero otherwise. This proves (14).

We next prove a deterministic bound for $S_{N,q}$. For $0 \leq m \leq q$, let

$$e_m(\varepsilon) = \sum_{\substack{I \subset [N] \\ |I|=m}} \prod_{x \in I} \varepsilon_x, \qquad e_0(\varepsilon) = 1. \tag{18}$$

The power sums of the Rademacher signs satisfy

$$p_j := \sum_{x=1}^{N} \varepsilon_x^j = \begin{cases} L_N, & j \text{ odd}, \\ N, & j \text{ even}. \end{cases} \tag{19}$$

Newton's identities state that

$$m e_m = \sum_{j=1}^{m} (-1)^{j-1} e_{m-j} p_j. \tag{20}$$

An induction on m, using (20), shows that there is a constant $C_m < \infty$ such that

$$|e_m(\varepsilon)| \le C_m \left(N^{m/2} + |L_N|^m \right). \tag{21}$$

For completeness, suppose that the estimate is known below order m and write $u = |L_N|/\sqrt{N}$. If j is even, then

$$|e_{m-j} p_j| \le C_{m-j} \left(N^{(m-j)/2} + |L_N|^{m-j} \right) N \tag{22}$$

$$\le C_{m-j} N^{m/2} \left(1 + u^m\right), \tag{23}$$

because $j \ge 2$ and $u^{m-j} \le 1 + u^m$. If j is odd, then

$$|e_{m-j} p_j| \le C_{m-j} \left(N^{(m-j)/2} + |L_N|^{m-j} \right) |L_N| \tag{24}$$

$$\le C_{m-j} N^{m/2} \left(1 + u^m\right), \tag{25}$$

because $j \ge 1$ and $u^{m-j+1} \le 1 + u^m$. Substitution into (20) proves the induction step. Since $S_{N,q} = e_q(\varepsilon)$, we have in particular

$$|S_{N,q}| \le C_q \left(N^{q/2} + |L_N|^q \right). \tag{26}$$

Set $X_N = L_N/\sqrt{N}$. Since $|X_N| \le \sqrt{N}$ and $a = q/2 - 1$, (26) implies that, for $0 \le t \le c_q N^{-(q-1)}$,

$$t|S_{N,q}| \le C_q c_q N^{-a} \left(1 + |X_N|^q\right) \tag{27}$$

$$\le C_q c_q \left(1 + |X_N|^2\right). \tag{28}$$

Indeed,

$$N^{-a} |X_N|^q = |X_N|^2 \left(\frac{|X_N|}{\sqrt{N}} \right)^{q-2} \le |X_N|^2. \tag{29}$$

Choose c_q small enough that the last member of (28) is bounded by $(1 + |X_N|^2)/4$.

Hoeffding's inequality gives

$$\mathbb{P}_\varepsilon(|X_N| \ge u) \le 2e^{-u^2/2} \qquad (u \ge 0). \tag{30}$$

Consequently, integration of the tail in (30) yields

$$\sup_{N \ge 1} \mathbb{E}_\varepsilon \left[\left(1 + |X_N|^{3q}\right) e^{|X_N|^2/4} \right] < \infty. \tag{31}$$

Finally, (26), (28), and (31) give

$$\mathbb{E}_\varepsilon \left[|S_{N,q}|^3 e^{t|S_{N,q}|} \right] \le C_q N^{3q/2} \mathbb{E}_\varepsilon \left[\left(1 + |X_N|^{3q}\right) e^{(1+|X_N|^2)/4} \right] \tag{32}$$

$$\le C_q N^{3q/2}, \tag{33}$$

which proves (15). □

Proposition 2.5. *There exists $\beta_{\mathrm{Rad}} = \beta_{\mathrm{Rad}}(q) > 0$ such that, for every fixed real β with $|\beta| < \beta_{\mathrm{Rad}}$,*

$$\sum_{r\geq 3} \|P_r Z_N(\beta)\|_2^2 = O\Big(N^{-3(q/2-1)}\Big) = o(N^{2-q}). \tag{34}$$

Proof. Put

$$t_N = \beta^2 \sigma_N^2 = \beta^2 (q-1)! N^{-(q-1)}. \tag{35}$$

The Hermite coefficient formula, Lemma 2.3, and $F_N(\beta) > 0$ give

$$\|P_r Z_N(\beta)\|_2^2 = \frac{1}{F_N(\beta)^2} \sum_{|\alpha|=r} \frac{1}{\alpha!} \left|\mathbb{E}\partial_g^\alpha Z_N(\beta)\right|^2 \leq t_N^r \sum_{\substack{|\alpha|=r \\ \alpha \text{ even}}} \frac{1}{\alpha!}. \tag{36}$$

Here the sum may be restricted to even α by the sign-gauge symmetry. By (14), summing (36) over all $r \geq 3$ yields

$$\sum_{r\geq 3} \|P_r Z_N(\beta)\|_2^2 \leq \sum_{r\geq 3} \frac{t_N^r}{r!} \mathbb{E}_\varepsilon\big[S_{N,q}^r\big] \tag{37}$$

$$\leq \mathbb{E}_\varepsilon\left[\sum_{r\geq 3} \frac{(t_N |S_{N,q}|)^r}{r!}\right] \tag{38}$$

$$\leq \frac{t_N^3}{6} \mathbb{E}_\varepsilon\left[|S_{N,q}|^3 e^{t_N |S_{N,q}|}\right]. \tag{39}$$

Choose

$$\beta_{\mathrm{Rad}} = \sqrt{\frac{c_q}{(q-1)!}}, \tag{40}$$

where c_q is the constant in Lemma 2.4. If $|\beta| < \beta_{\mathrm{Rad}}$, then $t_N \leq c_q N^{-(q-1)}$. Therefore (15) and (39) imply

$$\sum_{r\geq 3} \|P_r Z_N(\beta)\|_2^2 \leq C_{q,\beta} N^{-3(q-1)} N^{3q/2} \tag{41}$$

$$= C_{q,\beta} N^{-3(q/2-1)}. \tag{42}$$

Since $q \geq 4$, we have $3(q/2-1) > q-2$, which completes the proof. □

3 Proof of the main results

3.1 Annealed free energy and coefficient asymptotics

To identify the deterministic coefficient $A_N(\beta)$ in (3), we use the annealed free-energy limit from [5]. The required convergence of derivatives follows from convexity and is not an additional input.

Lemma 3.1. *Let $I \subset \mathbb{R}$ be open. Suppose that $h_N : I \to \mathbb{R}$ are finite, differentiable, convex functions and $h_N(x) \to h(x)$ pointwise, where $h \in C^1(I)$. Then $h_N \to h$ and $h_N' \to h'$ locally uniformly on I.*

Proof. Fix a compact interval $[a,b] \subset I$ and choose $u < a < b < v$ in I. Monotonicity of secant slopes bounds every secant slope of h_N on $[a,b]$ between quantities formed from $h_N(u), h_N(a), h_N(b), h_N(v)$. These endpoint values converge, so the functions h_N are equi-Lipschitz on $[a,b]$. Pointwise convergence on a finite mesh, followed by mesh refinement, gives uniform convergence there.

For $x \in I$ and sufficiently small $s > 0$, convexity gives

$$\frac{h_N(x) - h_N(x-s)}{s} \leq h_N'(x) \leq \frac{h_N(x+s) - h_N(x)}{s}.$$

First let $N \to \infty$ and then $s \downarrow 0$ to obtain $h_N'(x) \to h'(x)$. Finally, h_N' are nondecreasing. On a fine partition of $[a,b]$, monotonicity traps $h_N'(x)$ between its values at adjacent mesh points, while uniform continuity traps $h'(x)$ between the corresponding values of h'. Pointwise convergence at the finitely many mesh points then gives uniform convergence of the derivatives. □

Proposition 3.2. *There are $\beta_{\mathrm{FE}} = \beta_{\mathrm{FE}}(q) > 0$ and an even real-analytic function ℓ_q on $(-\beta_{\mathrm{FE}}, \beta_{\mathrm{FE}})$ such that, locally uniformly on this interval,*

$$\frac{1}{N} \log F_N(\beta) \longrightarrow \ell_q(\beta), \tag{43}$$

$$\frac{1}{N} F_N'(\beta) \longrightarrow \ell_q'(\beta). \tag{44}$$

Moreover,

$$\ell_q(\beta) = \frac{\beta^2}{2q} + O(\beta^4), \qquad \ell_q'(\beta) = \frac{\beta}{q} + O(\beta^3). \tag{45}$$

In paritcular, $\ell'(\beta) > 0$, for every sufficiently small real $\beta \neq 0$.

Proof. The pointwise convergence in (43) is Theorem 5 of [5], together with the Gaussian extension stated there. That extension invokes the Gaussian–Rademacher universality result [6].

We first record the normalization conversion. The Majoranas of [5] satisfy

$$\psi_i^{\mathrm{GPSZ}} = 2^{-1/2} \psi_i, \qquad H_N^{\mathrm{GPSZ}} = 2^{-q/2} H_N.$$

Set $u = 2^{q/2}\beta$. Gaussian symmetry gives

$$\mathbb{E} \operatorname{tr} e^{-\beta H_N} = \mathbb{E} \operatorname{tr} e^{+\beta H_N},$$

and $u H_N^{\mathrm{GPSZ}} = \beta H_N$. Thus the sign and normalization conventions of [5] give

$$\frac{1}{N} \log F_N(\beta) = \frac{u}{N} f_{N,q}^{\mathrm{GPSZ}}(u), \tag{46}$$

where $f_{N,q}^{\mathrm{GPSZ}}(u) = u^{-1} \log \mathbb{E} Z_N^{\mathrm{GPSZ}}(u)$.

We next extract the regularity and the first coefficient from the explicit formula in that theorem. In its notation,

$$\Phi_u(z) = \sum_{v \geq 1} \mu_v(u) z^v, \qquad u f_q^{\mathrm{GPSZ}}(u) = \Phi_u(t_*(u)) - 1 - \log t_*(u),$$

where $t_* \partial_z \Phi_u(t_*) = 1$. A contributing hypertree has $v = 1 + (q-1)s$ vertices and its coefficient contains

$$\rho(u)^{v-1}, \qquad \rho(u)^{q-1} = \frac{(q-1)! u^2}{2^{q+1}}.$$

More explicitly, $\mathcal{T}_v^{\mathrm{sim}}$ is empty unless $v = 1 + (q-1)s$ for some $s \geq 0$. For $s \geq 1$ we may therefore write

$$\mu_{1+(q-1)s}(u) = c_{q,s} u^{2s},$$

where

$$c_{q,s} = \frac{1}{(1+(q-1)s)!} \left(\frac{(q-1)!}{2^{q+1}} \right)^s \sum_{T \in \mathcal{T}_{1+(q-1)s}^{\mathrm{sim}}} b_T.$$

The bounds used in the proof of Proposition 3 of [5] give $|c_{q,s}| \leq C_q^s$. Hence

$$\Phi_u(z) = z + \sum_{s \geq 1} c_{q,s} u^{2s} z^{1+(q-1)s}$$

converges normally for (u^2, z) in a neighborhood of $(0,1)$. Thus $\Phi_u(z)$ is jointly analytic in (u^2, z) there. At $u = 0$, $\Phi_0(z) = z$ and $t_*(0) = 1$; the derivative with respect to t of $t\partial_z\Phi_u(t) - 1$ is 1 at $(u,t) = (0,1)$. The analytic implicit function theorem makes t_*, and hence the limiting right-hand side, analytic in u^2.

The first nontrivial term is the unique one-edge hypertree on q vertices:

$$\Phi_u(z) = z + \frac{u^2}{q2^{q+1}} z^q + O(u^4).$$

It follows successively that

$$t_*(u) = 1 - \frac{u^2}{2^{q+1}} + O(u^4), \qquad u f_q^{\mathrm{GPSZ}}(u) = \frac{u^2}{q2^{q+1}} + O(u^4).$$

Using $u = 2^{q/2}\beta$ in (46) gives (45).

It remains only to justify the two local-uniform assertions. Let $h_N = N^{-1} \log F_N$. If

$$\overline{\nu}_N(B) = \mathbb{E}\ \mathrm{tr}\, \mathbf{1}_{\{H_N \in B\}},$$

then $F_N(\beta) = \int e^{-\beta x} \overline{\nu}_N(\mathrm{d}x)$; hence h_N is finite, differentiable, and convex. Differentiation is justified by $\|H_N(G)\| \le \sigma_N \sum_{I \in \mathcal{I}_N} |g_I|$ and Gaussian exponential integrability. The limiting function just obtained is C^1. Lemma 3.1 now proves both local-uniform convergences. □

By (3) and (44),

$$\frac{A_N(\beta)}{N} \longrightarrow \frac{\beta}{2} \ell_q'(\beta). \tag{47}$$

Proof of Theorem 1.2. Let

$$Y_N(\beta; G) = \frac{Z_N(\beta; G)}{F_N(\beta)}, \qquad S_N = \frac{1}{\sqrt{2M_N}} \sum_{I \in \mathcal{I}_N} (g_I^2 - 1).$$

The classical i.i.d. central limit theorem gives $S_N \Rightarrow \xi \sim \mathcal{N}(0,1)$. By Lemma 2.2,

$$Y_N(\beta) - 1 = c_N(\beta) S_N + \mathcal{R}_N(\beta), \qquad c_N(\beta) = \frac{\beta\, F_N'(\beta)}{\sqrt{2M_N}}, \tag{48}$$

where $\mathcal{R}_N = \sum_{r \ge 3} P_r Z_N$ is orthogonal to S_N. Since $M_N \sim N^q/q!$, (44) gives

$$N^{q/2-1} c_N(\beta) \longrightarrow \sqrt{\frac{q!}{2}}\, \beta \ell_q'(\beta).$$

Proposition 2.5 gives $N^{q/2-1}\mathcal{R}_N \to 0$ in L^2. This proves (1). Orthogonality in (48) also gives

$$\mathrm{Var}(Y_N) = c_N(\beta)^2 + \|\mathcal{R}_N(\beta)\|_2^2 .$$

Finally, $\mathrm{Var}(Y_N) = \mathrm{Var}(Z_N)/(\mathbb{E}Z_N)^2$, and $\ell'(\beta) > 0$ for β small enough give (2). □

Lemma 3.3 (Gaussian concentration)**.** *Write*

$$W_N := \log \frac{Z_N(\beta)}{\mathbb{E}Z_N(\beta)}. \tag{49}$$

Then, for $\beta \neq 0$ and any $\lambda, t > 0$,

$$\mathbb{P}(W_N - \mathbb{E}W_N < -t) \le \exp\left(-\frac{qt^2}{2\beta^2}\right), \tag{50}$$

$$\mathbb{E}\exp(\lambda(W_N - \mathbb{E}W_N)) \le \exp\left(\frac{\lambda^2\beta^2}{2q}\right). \tag{51}$$

Proof. To apply the Gaussian concentration inequality, we view $Z_N(\beta; G)$ as a function of $G = (g_I)_{I \in \mathcal{I}_N}$, and establish its Lipschitz bound. Fix $I \in \mathcal{I}_N$. By Duhamel's formula, we deduce that

$$\begin{aligned} \partial_{g_I} Z_N &= \operatorname{tr} \partial_{g_I} e^{-\beta H_N(G)} && (52)\\ &= -\beta \operatorname{tr}\Big(\partial_{g_I} H_N(G) \cdot e^{-\beta H_N(G)} \Big) && (53)\\ &= -\beta \sigma_N \operatorname{tr}\big(e^{-\beta H_N} \psi_I \big). && (54) \end{aligned}$$

Write $\langle \cdot \rangle_G$ for the expectation with respect to the normalized Gibbs state, i.e.,

$$\langle A \rangle_G := \frac{\operatorname{tr}(A e^{-\beta H_N(G)})}{Z_N(\beta; G)}. \tag{55}$$

Then we obtain

$$\partial_{g_I} W_N = -\beta \sigma_N \langle \psi_I \rangle_G. \tag{56}$$

Fix an arbitrary subset $S \subseteq [N]$ with $|S| = q - 1$. Set $A_i := \psi_{S \cup \{i\}}$ for each $i \notin S$. Note that, by the relation $\{\psi_i, \psi_j\} = 2\delta_{ij}\mathbb{I}$, it holds that

$$A_i^2 = \mathbb{I}, \qquad A_i A_j + A_j A_i = 0 \quad \text{if} \quad i \neq j. \tag{57}$$

This yields

$$\left(\sum_{i \notin S} \langle A_i \rangle_G A_i \right)^2 = \sum_{i \notin S} \langle A_i \rangle_G^2 A_i^2 + \sum_{i,j \notin S i < j} \langle A_i \rangle_G \langle A_j \rangle_G (A_i A_j + A_j A_i) = \sum_{i \notin S} \langle A_i \rangle_G^2 \mathbb{I}. \tag{58}$$

If we write $\| \cdot \|$ for the operator norm, then, by (57), we obtain

$$\left\| \left(\sum_{i \notin S} \langle A_i \rangle_G A_i \right)^2 \right\| = \sum_{i \notin S} \langle A_i \rangle_G^2. \tag{59}$$

Since A_i is self-adjoint, it follows that

$$\left\| \sum_{i \notin S} \langle A_i \rangle_G A_i \right\| = \sqrt{\sum_{i \notin S} \langle A_i \rangle_G^2}. \tag{60}$$

On the other hand, Hölder's inequality for Schatten norms yields

$$\left\langle \sum_{i \notin S} \langle A_i \rangle_G A_i \right\rangle_G \leq \left\| \sum_{i \notin S} \langle A_i \rangle_G A_i \right\|. \tag{61}$$

Since the left-hand side equals $\sum_{i \notin S} \langle A_i \rangle_G^2$, we deduce from (60) and (61) that

$$\sum_{i \notin S} \langle A_i \rangle^2 \leq 1. \tag{62}$$

For each $I \in \mathcal{I}_N$, there exist exactly q subsets $S \subseteq [N]$ satisfying $I = S \cup \{i\}$. Hence,

$$\sum_{\substack{S \subseteq [N] \\ |S| = q-1}} \sum_{i \notin S} \langle \psi_{S \cup \{i\}} \rangle_G^2 = q \sum_{I \in \mathcal{I}_N} \langle \psi_I \rangle_G^2. \tag{63}$$

This, combined with (62), yields

$$\sum_{I \in \mathcal{I}_N} \langle \psi_I \rangle_G^2 \leq \frac{1}{q} \sum_{\substack{S \subseteq [N] \\ |S| = q-1}} 1 = \frac{1}{q} \binom{N}{q-1} \leq \frac{N^{q-1}}{q!}. \tag{64}$$

From this and (56), we deduce that

$$\|\nabla W_N\|_2^2 \le \frac{\beta^2 \sigma_N^2 N^{q-1}}{q!} = \frac{\beta^2}{q}, \tag{65}$$

where the last equality follows from the definition of σ_N^2. This implies that $W_N(G)$ is $|\beta|/\sqrt{q}$-Lipschitz as a function of $G = (g_I)_{I \in \mathcal{I}_N}$. Hence, the Gaussian concentration inequality gives the desired result. □

Proof of Corollary 1.3. The case $\beta = 0$ is trivial, so assume $\beta \neq 0$. Put

$$a = \frac{q}{2} - 1, \qquad X_N = \frac{Z_N(\beta) - \mathbb{E}Z_N(\beta)}{\mathbb{E}Z_N(\beta)}, \qquad W_N = \log Z_N(\beta) - \log \mathbb{E}Z_N(\beta).$$

It then holds that

$$\begin{aligned} \log Z_N - \mathbb{E}\log Z_N &= (\log Z_N - \log \mathbb{E}Z_N) + (\log \mathbb{E}Z_N - \mathbb{E}\log Z_N) && (66)\\ &= W_N - \mathbb{E}W_N && (67)\\ &= (W_N - X_N) + X_N - \mathbb{E}W_N, && (68) \end{aligned}$$

which yields

$$N^a(\log Z_N - \mathbb{E}\log Z_N) = N^a(W_N - X_N) + N^a X_N - N^a \mathbb{E}W_N. \tag{69}$$

By Theorem 1.2, we have

$$N^a X_N \Longrightarrow \sqrt{\frac{q!}{2}}\, \beta \ell_q'(\beta)\, \xi. \tag{70}$$

Let $C_0 \in (0, \infty)$ be such that $|\log(1+x) - x| \le C_0 x^2$ for all $x \in [-1/2, 1/2]$. Since $W_N = \log(1 + X_N)$, it follows that

$$\mathbb{E}\big[1 \wedge (N^a |W_N - X_N|)\big] \le \mathbb{E}\big[1 \wedge (C_0 N^a |X_N|^2)\big] + \mathbb{P}(|X_N| > 1/2). \tag{71}$$

By (70), $|X_N| \xrightarrow{\mathrm{p}} 0$ and $(N^a|X_N|)_{N \ge 1}$ is tight. Thus, the right-hand side of the above inequality converges to 0 as $N \to \infty$. Hence, $N^a(W_N - X_N) \xrightarrow{\mathrm{p}} 0$. Therefore, it suffices to show that $\mathbb{E}W_N = o(N^{-a})$.

We claim that

$$-\frac{\beta^2}{2q} \le \mathbb{E}W_N \le 0. \tag{72}$$

Indeed, the upper bound follows from Jensen's inequality, while the lower bound follows from (51) and the identity $\mathbb{E}e^{W_N} = 1$. Let $K_\beta = \beta^2/(2q)$ and $\delta_N := (\mathbb{E}X_N^2)^{1/2}$. By Theorem 1.2,

$$\delta_N^2 = \mathbb{E}X_N^2 \asymp N^{-(q-2)}. \tag{73}$$

Henceforth, we consider sufficiently large N so that $\delta_N \le e^{-1}$. Since $e^{W_N} - 1 = X_N$, Markov's inequality gives

$$\mathbb{P}(W_N < -t) \le \frac{\delta_N^2}{(1 - e^{-t})^2} \le C\delta_N^2, \qquad \forall t \ge 1. \tag{74}$$

On the other hand, by $\mathbb{E}W_N \ge -K_\beta$ and (50),

$$\mathbb{P}(W_N < -t) \le \exp\left(-\frac{q(t - K_\beta)^2}{2\beta^2}\right), \qquad \forall t > K_\beta. \tag{75}$$

Set

$$T_N = \max\left\{1,\ K_\beta + \frac{2|\beta|}{\sqrt{q}}\sqrt{\log(1/\delta_N)}\right\}.$$

By the layer-cake representation, we have

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] = \mathbb{P}(W_N < -1) + \int_1^\infty \mathbb{P}(W_N < -t)\,\mathrm{d}t.$$

Applying (74) to the first term and the integral over $[1, T_N]$ and applying (75) to the integral over (T_N, ∞), we obtain

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le CT_N\delta_N^2 + \int_{T_N}^\infty \exp\left(-\frac{q(t-K_\beta)^2}{2\beta^2}\right)\mathrm{d}t. \tag{76}$$

By the Gaussian tail bound $\int_a^\infty e^{-t^2}\mathrm{d}t \le Ce^{-a^2}$ for any $a \ge 0$, the second term is bounded by $C\delta_N^2$. Consequently,

$$\mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le C\delta_N^2\sqrt{\log(1/\delta_N)}. \tag{77}$$

Now set $h(w) = e^w - 1 - w$. Note that

$$h(w) \le C(e^w-1)^2 \quad (w \ge -1), \qquad h(w) \le |w| \quad (w < -1).$$

This, combined with $e^{W_N} - 1 = X_N$ and (77), yields

$$\mathbb{E}h(W_N) \le C\mathbb{E}X_N^2 + \mathbb{E}\big[|W_N|\,\mathbf{1}_{\{W_N<-1\}}\big] \le C\delta_N^2\sqrt{\log(1/\delta_N)}. \tag{78}$$

Moreover, since $h(W_N) = X_N - W_N$ and $\mathbb{E}X_N = 0$, we have $-\mathbb{E}W_N = \mathbb{E}h(W_N)$. Therefore, by (72), we deduce that

$$0 \le -\mathbb{E}W_N \le C\delta_N^2\sqrt{\log(1/\delta_N)}. \tag{79}$$

From this and (73), it follows that

$$\mathbb{E}W_N = O\Big(N^{2-q}\sqrt{\log N}\Big) = o(N^{-(q/2-1)}),$$

which completes the proof. □

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